SIMATS ENGINEERING

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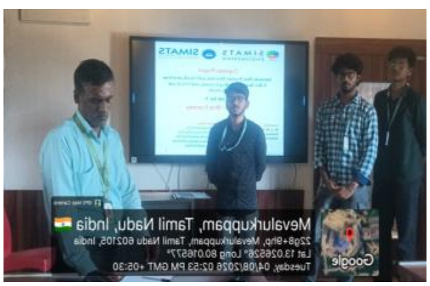
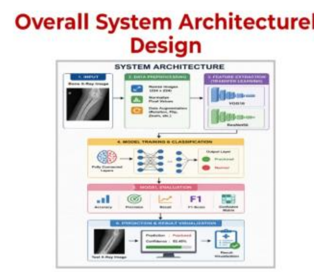
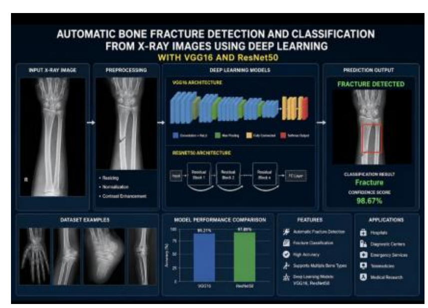
Course Code: MLA0405 **Slot**: D
Course Name: Deep Learning
Course Faculty: Dr. A.M.Arul Raj

Project Title: Automatic Bone Fracture Detection and Classification from X-Ray Images Using Deep Learning with VGG16 and ResNet50

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

Review 1 Image	**Review 2 Image**	**Module Image**
		

Student Contribution Module Work - Title	Demo Image
Module 3 – Bone Fracture Detection and Classification In this module, the features extracted from **VGG16 and ResNet50** are used for fracture detection. .The models are trained using the prepared X-ray training dataset	

Project Description: (here you write what you did in this project (contribution) including Model Description

Module 3 focuses on detecting and classifying bone fractures from X-ray images using the features extracted by the VGG16 and ResNet50 models. In this module, the preprocessed X-ray images and the important features obtained through Transfer Learning are given to the trained deep learning models. The models learn the differences between fractured and normal bones from the training data. After training, a Softmax Classifier is used to classify the input X-ray into two classes: “Fractured” or “Normal.” The trained models analyze important patterns such as edges, shapes, textures, and abnormalities in the bone region to make the classification. This module therefore converts the extracted image features into the final fracture detection result, providing an automated and consistent method for identifying bone fractures.

Guide Signature :   Student Signature :